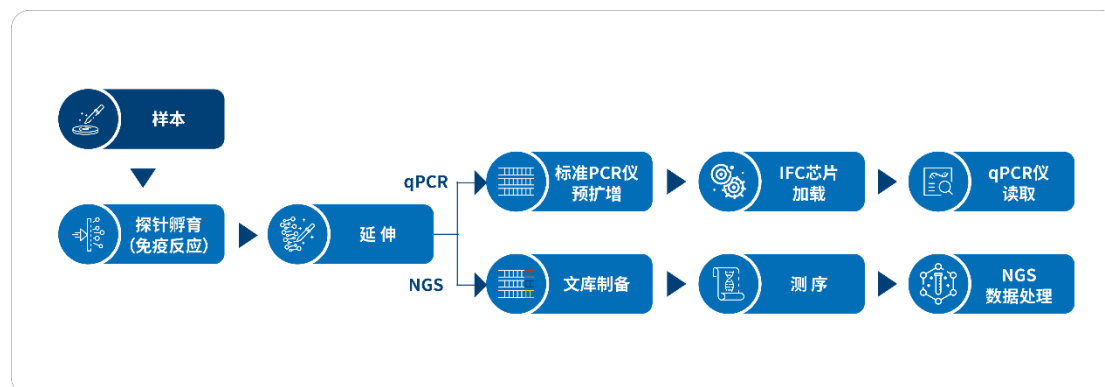


Method



1 Experimental Procedure

Samples and controls are transferred to a sample plate. High-multiplex matched pairs of antibodies, labeled with unique DNA oligonucleotides, bind to their respective proteins in the samples. Proximal oligonucleotides hybridize and undergo extension by Polymerase Chain Reaction (PCR). Final DNA amplicons include specific barcode sequences for each assay and sample specific indexes, which are then detected on an NGS sequencer or a Signature Q100.



2 Bioinformatics Analysis Pipeline

2.1 Data Normalization and Quality Control

NPX, Normalized Protein eXpression, is Olink's arbitrary unit which is in Log2 scale. Following sequencing, the raw data is converted to counts, assigning an integer value to each assay-sample combination based on the detected copy numbers. These raw data counts are converted into NPX values, enabling the identification of protein level changes within different sample sets, and establishing protein signatures. In general, a higher NPX value refer to a higher protein concentration.

2.2 Differential Expression Analysis

The protein expression difference is calculated using standardized NPX data, employing the t-test or ANOVA method under different conditions. The t-test is used for comparing a single variable between two groups, while ANOVA is employed when covariates exist, and more than two groups are being compared. For each method, the analysis is performed separately using the function `olink_ttest` and `olink_anova` from the R package OlinkAnalyze (Olink Proteomics AB, 2023). Significant differentially expressed proteins are then filtered using adjusted p-values.

2.2.1 `olink_ttest`

`olink_ttest` performs a t-test for a given grouping variable as defined by variable argument. Paired t-test can be performed by defining `pair_id` argument. Variable must have 2 groups with more than 2 samples per group.

2.2.2 olink_anova

olink_anova performs an ANOVA F-test for each protein based on variables and covariates defined by variable and covariate arguments or by formula defined by model_formula argument.

2.3 Enrichment Analysis

Gene Ontology (GO) enrichment analysis of differentially expressed genes were implemented by the clusterProfiler R package, in which gene length bias was corrected. GO terms with corrected P-value less than 0.05 were considered significantly enriched by differentially expressed genes. KEGG is a database resource for understanding high-level functions and utilities of the biological system, such as the cell, the organism and the ecosystem, from molecular-level information, especially large-scale molecular datasets generated by genome sequencing and other high-through put experimental technologies (<http://www.genome.jp/kegg/>). We used clusterProfiler R package to test the statistical enrichment of differential expression genes in KEGG [1] pathways. The Reactome database brings together the various reactions and biological pathways of human model species. Reactome pathways with corrected P-value less than 0.05 were considered significantly enriched by differential expressed genes.

3 Reference

[1] Kanehisa M, Goto S. KEGG: kyoto encyclopedia of genes and genomes[J]. Nucleic acids research, 2000, 28(1): 27-30.

Suggested Materials & Methods text for describing Olink in publications

1、Reveal/Explore HT/ Explore 3072/ Explore 1536/ Explore 384

Proteins were measured using Olink® [state Explore 384, Explore1536, Explore 3072, Explore HT or Reveal used] (Olink Proteomics AB, Uppsala, Sweden) according to the manufacturer's instructions. The technology behind the Olink protocol is based on Proximity Extension Assay (PEA) (Assarsson et al, 2014), coupled with readout via nextgeneration sequencing (NGS). In brief, pairs of oligonucleotide-labeled antibody probes designed for each protein bind to their target, bringing the complementary oligonucleotides in close proximity and allowing for their hybridization. The addition of a DNA polymerase leads to the extension of the hybridized oligonucleotides, generating a unique protein identification “barcode”. Next, library preparation adds sample identification indexes and the required nucleotides for sequencing. Prior to sequencing using the Illumina® NovaSeq™6000, libraries go through a bead-based purification step and the quality is assessed using the LabChip GX Touch (Revvity Health Sciences, Inc., Waltham, MA). The raw output data is quality controlled, normalized and converted into NPX®, Olink’s proprietary unit of relative abundance. Data normalization is performed using an internal extension control and an external plate control to adjust for intra- and inter-run variation. All assay validation data (analytical range, intra-plate and inter-plate assay precision data etc.) are available on manufacturer's website (www.olink.com).

2、Target 48/96

Proteins were measured using the Olink® XX panel* (Olink Proteomics AB, Uppsala, Sweden) according to the manufacturer's instructions. The Proximity Extension Assay (PEA) technology used for the Olink protocol has been well described (Assarsson et al, 2014). In brief, pairs of oligonucleotide-labeled antibody probes bind to their targeted protein, and if the two probes are brought in close proximity the oligonucleotides will hybridize in a pair-wise manner. The addition of a DNA polymerase leads to a proximity-dependent DNA polymerization event, generating a unique PCR target sequence. The resulting DNA sequence is subsequently detected and quantified using a microfluidic real-time PCR instrument (Signature Q100). Data is then quality controlled and normalized using an internal extension control and an inter-plate control, to adjust for intra- and inter-run variation. The final assay read-out is presented in Normalized Protein eXpression (NPX) values, which is an arbitrary unit on a log2-scale where a high value corresponds to a higher protein expression. All assay validation data (detection limits, intra- and inter-assay precision data, etc.) are available on manufacturer's website (www.olink.com).